Learning Objective 1.3: I can determine the polarization of an antenna and describe the bandwidth characteristics of common antenna types.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Identify the polarization.** For each plane wave propagating in $+\hat{\mathbf{z}}$, identify the polarization (linear, RHCP, LHCP, or elliptical) and — where linear — the tilt angle ψ measured from $\hat{\mathbf{x}}$.

   (a) $\vec{E} = \hat{\mathbf{x}}3\cos(\omega t - kz)$

   (b) $\vec{E} = \hat{\mathbf{x}}2\cos(\omega t - kz) + \hat{\mathbf{y}}2\cos(\omega t - kz)$

   (c) $\vec{E} = \hat{\mathbf{x}}5\cos(\omega t - kz) + \hat{\mathbf{y}}5\cos(\omega t - kz - 90^\circ)$

   (d) $\vec{E} = \hat{\mathbf{x}}5\cos(\omega t - kz) + \hat{\mathbf{y}}5\cos(\omega t - kz + 90^\circ)$

   (e) $\vec{E} = \hat{\mathbf{x}}3\cos(\omega t - kz) + \hat{\mathbf{y}}1\cos(\omega t - kz - 90^\circ)$

2. **Axial ratio.** A satellite downlink antenna is spec'd circularly polarized with AR $\leqslant 3$ dB across its operating band.

   (a) Convert the 3 dB axial ratio to a linear ratio (major/minor axis).

   AR = $\boxed{}$

   (b) Write the two orthogonal linear components (up to a common scale) with the appropriate 90° phase, assuming the ellipse major axis is along $\hat{\mathbf{x}}$.

   (c) An $\hat{\mathbf{x}}$ receiver picks off E_x; a $\hat{\mathbf{y}}$ receiver picks off E_y. Compute the ratio of received *power* in dB. Should it match part (a)?

   $P_x/P_y =$ $\boxed{}$

3. **Polarization loss factor.**

(a) A ground station transmits a *linearly* polarized signal along $\hat{\mathbf{x}}$. A satellite receiver is **RHCP** and pointed correctly. Compute the PLF in dB.

PLF = $\boxed{}$

(b) Same ground station, but the satellite receiver is **LHCP**. Compute the PLF.

PLF = $\boxed{}$

(c) Two linear whip antennas: radio A vertical, radio B tilted $\psi = 30^\circ$ from vertical. Compute the PLF.

PLF = $\boxed{}$

(d) In one sentence, explain why aircraft-to-ground links almost always specify circular polarization on at least one end.

4. **Impedance bandwidth from a VSWR spec.** An antenna is spec'd "VSWR $\leqslant 2$ from 2.30GHz to 2.55GHz."

   (a) What is the impedance bandwidth in MHz?

   BW = $\boxed{}$

   (b) What is the fractional bandwidth (FBW) in percent?

   FBW = $\boxed{}$

   (c) Convert VSWR = 2 to $|\Gamma|$ and to return loss in dB.

   $|\Gamma|$ = $\boxed{}$

   RL = $\boxed{}$

   (d) Narrowband, broadband, or ultra-wideband? Name the likely antenna family, using the bandwidth table from Lesson 3.

5. **Fractional and ratio bandwidth.**

(a) A cellular antenna covers 698–960MHz (low) and 1.71–2.69GHz (high). Compute the FBW and RBW of each band.

low =

high =

(b) A UWB radar antenna covers 3.1–10.6GHz. Compute its RBW. Does it meet the FCC UWB threshold of RBW $\geqslant$ 2:1?

RBW =

(c) Which antenna family would you pick for each of the three bands? Give a one-sentence reason for each that names the *mechanism* buying the bandwidth — resonant, traveling-wave, or self-scaling.

6. **Chu-Harrington intuition.** A design must fit inside a sphere of radius $a = 15$ mm and operate at $f = 915$MHz (ISM band).

   (a) Compute λ, k, and the electrical size ka.

   $\lambda =$

   $k =$

   $ka =$

   (b) Estimate the minimum Q from $Q_{\text{min}} \approx \dfrac{1}{(ka)^3} + \dfrac{1}{ka}$.

   $Q_{\text{min}} =$

   (c) Estimate the maximum fractional bandwidth $\text{FBW}_{\text{max}} \approx 1/Q_{\text{min}}$ (for VSWR $\leqslant 2$).

   $\text{FBW}_{\text{max}} =$

   (d) The designer wants 100 MHz of usable bandwidth at 915 MHz. Possible in a 15 mm sphere?

7. **Reading the datasheet.** A vendor offers a GPS patch specified as **RHCP**, AR $\leqslant 3$ dB , "bandwidth 4% (VSWR $\leqslant 2$)". You are choosing the antenna for the other end of the link.

   (a) The datasheet defines handedness *from the receiver looking back toward the incoming wave.* Which convention is that, what is the part's sense in **IEEE** terms, and what goes wrong if you pair it with a satellite transmitting IEEE RHCP?

   (b) Convert AR $= 3$ dB to a linear ratio A, and give A^2.

   $A =$ ☐

   (c) A *linear* antenna receives from this patch. Using $\mathrm{PLF}(\psi) = \dfrac{A^2\cos^2\psi + \sin^2\psi}{A^2+1}$, find the best and worst PLF over all orientations ψ, in dB.

   best = ☐

   worst = ☐

   (d) Show that the peak-to-null swing in dB equals the axial ratio in dB.

(e) The far end is an opposite-sense CP antenna, also AR = 3 dB , major axes aligned. Worst-case rejection is $\left(\frac{A^2-1}{A^2+1}\right)^2$. Evaluate it and compare with the idealized PLF table.

rejection =

(f) Name two things you must ask before believing the antenna works over the advertised 4%.

Documentation: ______________________________